

Subhan Sajadi

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Education

BEng, Mechanical Engineering (Mechatronics Option), Minor in Computer Science Sep 2022 – Apr 2027

Toronto Metropolitan University • Ontario Scholar • Entrance Scholarship

Coursework: Fundamentals of Robotics, Measurements, Sensors & Instruments, Manufacturing System Control, Vibrations, Microprocessor Systems, Fluid Mechanics, Heat Transfer, Manufacturing Fundamentals

Research Experience

SymBioRobotX Lab, Toronto Metropolitan University

Toronto, ON

Undergraduate Research Assistant

Jul 2026 – Present

- Built a **MATLAB/Simulink Desktop Real-Time** controller for a CubeMars AK70-9 exosuit actuator over **CAN** (PCAN-USB, 1 Mbps), with a position/velocity/current mode selector, a safe-disable path, and live commanded-vs-measured plotting.
- Moved the controller to real-time kernel mode, eliminating a ~4 s feedback lag, then measured ~80% amplitude tracking at 1 Hz and CAN timing jitter to characterize bandwidth for gait-phase assistance.
- Debugged the CAN stack by hand-decoding frames in PCAN-View, tracing silent driver clamping to a 10,000× command-scaling error and a misassigned current-loop ID, bringing current control online.
- Designed a 12-position CAN/48 V power tether (Amphenol AT-series) with a pin map, bus termination, and a mandatory pre-connection isolation test; replacing 10–14-week-lead M23 connectors with stock parts cut the BOM to ~US\$233.
- Developed an 8-stage actuator bench test plan with the PI (identification through torque, thermal, and fault tests) and designed a 3D-printed motor testbed with a lever arm for gravity-referenced torque calibration.

Miniaturization in Bioanalysis Lab, iBEST (St. Michael's Hospital)

Toronto, ON

Undergraduate Research Assistant

May 2026 – Aug 2026

- Developed a two-layer **SU-8 photolithography** process on 100 mm silicon wafers using a MicroWriter ML3 maskless direct-write system to fabricate mother-machine microfluidic devices for single-cell *E. coli* imaging.
- Fabricated PDMS devices end to end (casting, port punching, plasma bonding) and iterated photomask geometry across multiple revisions, writing a **Python** GDS viewer to inspect mask files.
- Diagnosed resist-thickness discrepancies by cross-referencing optical profilometry against spin-speed parameters, and designed discriminating experiments to separate instrument artifacts from coating defects.
- Authored the lab's 30-section cleanroom SOP (v3.0) for the full two-layer workflow with alignment-mark recovery fallbacks, and designed a 3D-printed microscope stage insert in **SolidWorks** for handoff to future students.

Professional Experience

Litens Automotive Group

Vaughan, ON

Purchasing Student

May 2025 – May 2026

- Built an **Excel/Power Query** cost-benchmarking platform comparing supplier pricing by part type, material, vendor, currency, and historical trend.
- Supported supplier cost forecasting and negotiations contributing to projected multi-million-dollar savings on future automotive programs.
- Evaluated supplier quotes using engineering drawings, BOMs, and cost drivers across machining, forging, casting, and injection moulding; drawing and BOM reviews identified design improvements worth up to 10% in savings.
- Automated purchasing workflows in **VBA**, saving up to 2 hours/day with zero errors on automated POs; managed ~200 RFQs and ~200 BOM action files and built ERP/PLM training resources.

Celestica

Toronto, ON

Commodity Management

May 2024 – Aug 2024

- Owned price determination, sourcing, and supplier negotiations for assigned commodities, achieving a 10% increase in cost savings.
- Analyzed BOMs and maintained material price databases to prepare competitive, compliant customer bids, and resolved supplier performance issues through quarterly business reviews.

Technical Projects & Design Teams

Computer Vision Developer, TMU Wave Autonomous Underwater Vehicle Team

- Developing **OpenCV** detectors for gates, slalom poles, and path markers for an autonomous underwater vehicle, running in a Dockerized ArduPilot SITL and MAVLink simulation stack.
- Proposed a shared detection-output contract (found, center, offset, direction, confidence) and a reusable center-offset utility to simplify integration with the controls and state-machine sub-teams.

1/10-Scale Autonomous Race Car, TMU Velocity (F1Tenth)

- Implemented SLAM-based localization with RViz visualization in **ROS2**, reaching sub-100 ms perception-to-control latency.
- Engineered LiDAR emergency braking with time-to-collision logic, cutting collision events by ~90% in testing.

Smart Rehabilitation Knee Sleeve, True North Biomedical Competition

- Designed and prototyped a smart knee sleeve integrating surface **EMG** and **IMU** sensors to quantify muscle-activation asymmetry and knee range of motion, targeting objective ACL rehabilitation metrics.
- Modelled the full CAD assembly and exploded BOM, with modular sensor mounts, a battery housing, and strain-relieved wire routing for repeated knee flexion.
- Completed a Health Canada/FDA Class I regulatory analysis, specifying electrical isolation and biocompatible materials.

Rocker-Bogie Competition Rover, Take-Off Robotics Competition

- Built a 6-wheel rocker-bogie rover with a servo-driven 3-DOF arm and claw for a three-event competition (obstacle time trials, block placement, laser targeting) on a 5-person team.
- Integrated an adapted rocker-bogie suspension with a custom chassis and arm in **SolidWorks**, and modelled arm workspace, worst-case torque, and tip-over stability in **MATLAB**.
- Programmed **ESP32** PWM motor and servo control with wireless gamepad input; 3D printed and assembled the rover.

Passive Vibration Absorber for a Helicopter Tail Rotor, Vibrations Design Project

- Designed a 2-DOF passive absorber for a 500 g unbalanced rotor mass at 1000–2000 rpm, tuning mass, damping, and frequency ratios with Den Hartog's equal-peak method.
- Reduced peak transmissibility by up to 80% at resonance, verified with **MATLAB** simulations of force transmissibility and amplitude response.

Robotic Gripper Mechanism, Mechanism Design Project

- Designed a 5-bar, 2-DOF gripper driven by dual servos synchronized through gears, with kinematic, static, and torque analysis for a secure grip on objects up to 500 g.
- Simulated motion and workspace in **MATLAB**, modelled the assembly in SolidWorks, and 3D printed a functional prototype.

EMG Signal Processing Pipeline, Biosignals Project

- Built a **MATLAB** pipeline (20–450 Hz bandpass, 60 Hz notch, rectification, 100 ms RMS envelope) normalizing tibialis anterior EMG to %MVC.

Leadership

VP Competitions, Ontario Engineering Competition 2027

Jan 2026 – Present

- Co-lead 9 provincial competitions for 300+ Ontario students, directly overseeing 6 competitions and ~20 leads.
- Led the 2026–2027 rulebook and AI-use policy revision, oversaw redesigns of two design challenges after test day exposed flaws, and manage a \$5K build budget.
- Built OEC Tracker (**React**, **Supabase**, **Node**), which polls meeting recordings every 15 min, extracts action items with an LLM (~\$0.01/meeting) behind a human review gate, and posts overdue reminders to Discord.
- Prototyped a **FastAPI**/SQLite judging portal with role-based logins, rubric scoring, timers, and CSV export.

VP Events, Mechatronics Course Union

May 2025 – Apr 2026

- Led 4 directors and 6 first-year reps, running 20+ technical events (SolidWorks, Arduino, Raspberry Pi, ROS2, AI/ML) for 150+ engineering students.
- Built the union's events dashboard (React/TypeScript, Supabase with row-level security, Netlify) with a daily GitHub Actions job that ingests union Instagram posts into a cross-union calendar.

VP Competitions, Metropolitan Engineering Competition

May 2023 – Nov 2025

- Ran 9 technical competitions per year across two cycles (18 total, ~16 teams each, ~400 competitors in 2024), leading teams of 11 and then 18 engineers and managing ~\$2K in materials procurement.
- Helped implement judging-scorecard automation (Google Apps Script, Python) and an inventory website with ~500 users.

Skills

Mechanical: SolidWorks, Fusion 360, Onshape, AutoCAD, ANSYS FEA, GD&T, Engineering Drawings, Metrology, CMM

Electrical & Embedded: ESP32, Arduino, CAN Bus (PCAN), I2C/SPI, KiCad, FPGA/VHDL

Controls & Analysis: MATLAB, Simulink, Simulink Desktop Real-Time, Signal Processing, Statistical Analysis

Software: Python, C/C++, Java, TypeScript/React, SQL, VBA, Bash, Git, Docker

Robotics & Vision: ROS2, RViz, OpenCV, YOLOv8, ArduPilot SITL, MAVLink

Fabrication & Lab: 3D Printing (FDM), CNC Machining, SU-8 Photolithography, PDMS Soft Lithography, Cleanroom Protocols

Business Tools: Excel/Power Query, Power BI, SAP/ERP, PLM